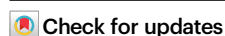


Atomic-resolution imaging reveals nucleus-free crystallization in two-dimensional amorphous ice on graphite

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Two-dimensional (2D) crystallization at interfaces or in thin films plays a critical role in many natural phenomena and technological applications, yet its microscopic mechanism remains elusive due to the challenges of directly observing atomic-scale transient states during crystallization. Here, we present atomic-resolution imaging of 2D ice crystallization on graphite surface using qPlus-based cryogenic atomic force microscopy (AFM) combined with molecular dynamics (MD) simulations. The crystallization of 2D amorphous bilayer ice undergoes a fractal-to-compact transition as temperature increases. Instead of forming a critical nucleus as predicted by classical theories, the crystallization firstly proceeds via the dendritic extension of fractal islands, followed by compact growth with defect healing at the percolated edges. We find that this process is significantly assisted by out-of-plane adsorbed (ad-) water molecules, which, like a spider weaving its web, facilitate the rearrangement of hydrogen-bonding network from disordered pentagons or heptagons to ordered hexagons. This fractal-to-compact crystallization pathway, mediated by ad-molecules, presents a non-classical ordering mechanism beyond classical nucleation theory, and may offer general insights into the crystallization at the 2D limit.

Crystallization is a fundamental process in materials science and natural systems, underpinning phenomena ranging from mineral formation to the fabrication of functional materials. In three-dimensional (3D) bulk systems, crystallization is commonly described by classical nucleation theory (CNT)¹, which involves the formation and growth of a critical nucleus^{2–4}. However, when crystallization occurs in two-dimensional (2D) or interfacial environments, the process becomes more complex and may deviate from this classical picture. Studies using 2D model systems—such as

colloidal suspensions, plasma crystals, and confined soft matter—have revealed complex features including transient metastable phases^{5,6} and anisotropic crystallites with rough, fractal-like boundaries^{7–9}, implying non-classical crystallization pathways. In molecular systems approaching the 2D limit, our understanding is even more limited, primarily due to the experimental challenge of resolving transient atomic-scale structures during nucleation and growth^{4,10}. Gaining insight into the microscopic mechanisms of 2D crystallization is therefore critical—not only for testing the limits of

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CNT, but also for enabling the rational design of low-dimensional materials with tailor-made properties^{11–15}.

A significant obstacle to elucidating the microscopic crystallization pathway lies in the experimental challenge of capturing the morphology of amorphous phases and their transient evolution with atomic-level resolution. Conventional diffraction and scattering techniques^{16–18} provide statistical structural information averaged over macroscopic regions, but lack the ability to directly image atomic configurations in real space. High-resolution transmission electron microscopy (HR-TEM) has enabled visualization of crystalline domains and even growing nuclei at atomic resolution^{3,4,13}. However, for disordered systems, its ability to resolve 3D amorphous structure is limited by the 2D projection of complex atomic arrangements^{19,20}. Nevertheless, pioneering studies have used TEM to investigate vitreous structures in monolayer silica^{21,22} and carbon-based 2D materials^{12,23}. Yet, the high-energy electron beams required for such imaging can damage fragile atomic bonding networks^{24,25}—particularly at crystal front edges²⁶, dislocations²⁷, and other structural defects^{28,29}—thereby impeding the accurate identification of intermediate structures during crystallization. As a result, the microscopic mechanism of atomic rearrangement and crystallization pathway remains poorly understood, leaving the debate between classical and non-classical models unresolved at the atomic level.

Using qPlus-based^{25,30–32} atomic force microscopy (AFM) and molecular dynamics (MD) simulations, we reveal the atomic-scale crystallization mechanism of 2D bilayer ice on a graphite substrate. The 2D ice islands undergo a morphological transition from fractal to compact with increasing temperature, revealing a non-classical, two-step crystallization pathway in the absence of critical nucleus formation. Strikingly, the structural ordering of 2D bilayer ice is not restricted to in-plane atomic rearrangements but is significantly influenced by out-of-plane adsorbed (ad-) water molecules. These ad-molecules dynamically shuttle between the adsorbate layer and the 2D bilayer structure, mediating 3D interactions vital for the 2D crystallization process. This study offers fundamental insights into low-dimensional crystallization pathways at atomic resolution and

provides guidance for the precise control of structural morphology in 2D materials³³.

Results

Phase diagram

The 2D amorphous ice was prepared by water vapor deposition onto a graphite surface at 15 K under ultra-high vacuum conditions, followed by annealing at various elevated temperatures (T_{exp}) to induce crystallization. In contrast to previous studies on metallic substrates^{25,34}, where interfacial ice morphology is strongly governed by surface atomic structures, the graphite substrate was chosen for its weak interaction and structural incommensurability with water molecules^{35,36}, enabling minimal structural templating and preserving metastable states during the crystallization process (see Supplementary Figs. 1, 2). These evolving ice morphologies are directly captured using AFM (Fig. 1a–c, see Methods and Supplementary Figs. 3, 4 for details), revealing the emergence of bilayer hexagonal ice (BHI) from the initial 2D amorphous structures.

Interestingly, the BHI islands (marked by yellow disks in Fig. 1a–f) exhibit three distinct phases with increasing annealing temperatures. At low temperatures (70 K in Fig. 1a, see also Supplementary Fig. 5), small dendritic hexagonal ice islands form, characterized by narrow branches approximately one hexagon in width, embedded within a matrix of 5-, 7-, and 8-membered rings typical of 2D amorphous ice³⁷. Radial distribution analysis (see Supplementary Fig. 6) confirms that these interlocked bilayer ring structures originate from the intrinsic dynamics of 2D ice, rather than substrate-induced effects. Upon annealing to 95 K (Fig. 1b), the hexagonal ice islands grow in both size and branch length, while maintaining their dendritic morphology. At higher temperatures (120 K; Fig. 1c), the islands grow into larger, more compact hexagonal structures, separated by line defects comprising alternative pentagonal and heptagonal rings. These three morphologies are also observed in MD simulations at corresponding annealing temperatures T_{sim} (Fig. 1d–f and Methods), in close agreement with the AFM observations. The hydrogen-bonding network topology is analyzed by connecting the neighboring water molecules within a cutoff

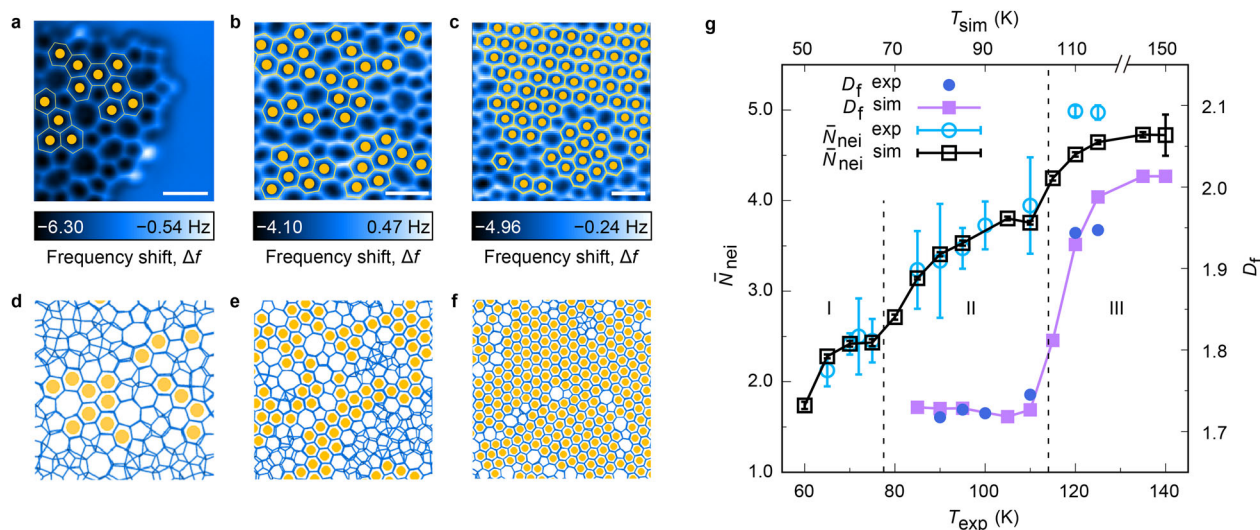


Fig. 1 | Phase diagram of bilayer hexagonal ice on graphite substrate. a–c Constant-height AFM (Δf) images of two-dimensional (2D) bilayer ice on graphite surface. The ice islands of interlocked bilayer hexagons are marked in yellow disks, displaying three typical phases with increasing annealing temperature: (I) small dendritic 2D ice islands with thin arms (**a**), (II) extended dendritic islands with thick arms (**b**) and (III) compact islands separated by grain boundaries of 5–7-membered rings (**c**). All the scale bars are 1 nm in length. **d–f**. Molecular dynamics snapshots of identical three phases at $T_{\text{sim}} = 55$ K (**d**), 100 K (**e**), 150 K (**f**). **g**. Temperature-

dependent phase diagram of 2D bilayer ice islands. The evolution of fractal dimension D_f and average number of neighboring hexagons $\bar{N}_{\text{nei}}$ with increasing temperature are plotted by the filled and hollow symbols respectively, exhibiting good agreement between experiment (dark and light blue circles) and simulation (connected purple and black squares) result. The error bar represents the standard deviation of $\bar{N}_{\text{nei}}$ derived from three independent measurements. The vertical black dashed lines represent the phase boundary between I, II, and III.

radius of $r_c = 3.5\text{\AA}$ (see Supplementary Fig. 2c), enabling clear identification of the ring structures in different crystallization regimes.

The temperature-dependent phase diagram of the three phases is summarized in Fig. 1g. The boundaries between phases I, II, and III can be quantitatively defined using the $\bar{N}_{\text{nei}}$ order parameter, which represents the average number of neighboring hexagons around a central hexagon within the BHI islands. This parameter ranges from 0 for isolated single hexagons to 6 for a defect-free infinite monocrystal (see Supplementary Fig. 7c), reflecting the compactness of the BHI islands. Notably, $\bar{N}_{\text{nei}}$ increases with temperature and exhibits two inflection points at $\bar{N}_{\text{nei}} \approx 2.5$ and 4.0 (vertical black dashes in Fig. 1g), marking structural transitions at $T_{\text{exp}} = 80 \pm 5\text{ K}$ ($T_{\text{sim}} = 70 \pm 5\text{ K}$) and $115 \pm 5\text{ K}$ ($105 \pm 5\text{ K}$) in both experiments and simulations. To further characterize the morphological evolution, we analyze the fractal dimension D_f of BHI islands, defined by the dimensional scaling relationship³⁸ as $\mathcal{N} \propto l^{D_f}$. Here $\mathcal{N}$ is the number of measurement units (e.g., rulers or squares) covering the measured object, and l is the scaling factor. In the context of our 2D system, $\mathcal{N}$ corresponds to the BHI island area A , and l is the characteristic island length³⁹ (see Supplementary Fig. 7a,b). As shown in Fig. 1g (filled circles and squares), the fractal dimension exhibits a discontinuous increase from $D_f \approx 1.7$ to ≈ 2.0 between phase II and III, confirming a temperature driven transition from a fractal to compact morphology in the BHI islands.

Structure and dynamic insight of ad-molecules

To gain deeper insight into the structural evolution during the crystallization process, we developed a fast-cooling sample preparation technique (see Methods) that effectively preserves intermediate states during crystallization. Using this technique, we observed the presence of additional water molecules adsorbed above the 2D bilayer ice, forming monomer and dimer structures particularly at the growth front edges of hexagonal ice (highlighted by green circles in Fig. 2a–j, see also Supplementary Fig. 8 for different tip-height images). These AFM images were analyzed and interpreted into atomic structures using a machine learning (ML) aided strategy⁴⁰, which was further validated by density functional theory (DFT) calculations and AFM simulations. The presence of adsorbed monomer and dimer structures is also observed extensively in MD simulations (Supplementary Figs. 9, 10). Unlike the interlocked hexagons in BHI or the 5- and 7-membered rings in 2D amorphous ice, the ad-water molecules are not interlocked with bottom-layer molecules. Instead, they form more intricate multi-layered structures with lower binding energy (see Supplementary Fig. 11), which may facilitate the breakage and reorganization of interlocked hydrogen bonds during the crystallization process.

The microscopic structures of the ad-molecules are further explored through MD simulations. These micro-structures centered on ad-molecules undergo a structural transformation as the system transitions between fractal (I & II) and compact (III) phases, as marked by the vertical black dash in Fig. 2k,l in accordance with the phase diagram (Fig. 1g). This transition can be characterized by the probability p_{low} of finding an underlying molecule directly beneath the ad-molecule (see Supplementary Figs. 9, 10). In the low temperature regime of phases I and II, few molecules are found beneath the monomer or dimer, with $p_{\text{low}} \approx 0$. As the temperature increases and the system enters phase III, a considerable number of molecules appear beneath the ad-molecules, marked by a jump discontinuity in p_{low} for both monomer and dimer structures. This structural transformation suggests that, at higher temperatures, the interplay between ad-molecules and 2D bilayer molecules becomes increasingly pronounced, facilitating the cooperative healing of interlocked pentagon and heptagon ring defects.

The dynamic behavior⁴¹ of the ad-molecules is also studied through MD simulations to uncover its correlation with the macroscopic phase diagram at different temperatures. The dynamics of the ad-molecules is characterized by their residence time τ , defined as the

reciprocal of hopping frequency between the 3D adsorbates and the 2D ice layers (see Methods and Supplementary Fig. 12). As shown in Fig. 2m, τ decreases with increasing temperature, exhibiting two dynamic crossover points that coincide with the transition boundaries between the I, II and III phases in the phase diagram. This suggests that the dynamic behavior of the ad-molecules plays a crucial role in the phase transition and crystallization process of the 2D bilayer ice.

Microscopic ordering pathway at dendrite tip and zigzag edge

To validate the ad-molecule-assisted 2D growth mechanism, we investigate the microscopic ordering pathways at the BHI growth front using MD simulations. Specifically, we focus on the distinct pathways of fractal growth at the dendrite tip and compact growth at zigzag edges, aiming to unveil the microscopic origins of the fractal structures formed at low temperatures and compact structures at high temperatures. Given the complexity of the multi-layered amorphous structures, an unsupervised ML strategy⁴¹ is employed to classify the various micro-structures at the BHI growth front (see Methods and Supplementary Fig. 13 for details). This algorithm efficiently classifies the local structures into several categories based on two principal components, PC1 and PC2, which correspond to the 5-, 6-, and 7-membered rings as well as other amorphous configurations (Fig. 3d, i). Transitions between these clusters represent intermediate states during structural conversion, allowing us to visualize the microscopic crystallization pathways⁴². In particular, we study the ordering pathway from pentagonal to hexagonal rings at the dendrite tip and zigzag edge, as highlighted by the white boxes in Fig. 3d, i and shown in Fig. 3a–c, f–h, respectively. Additional pathways for other structures are presented in Supplementary Fig. 14.

Figure 3a–c illustrates the dendrite tip growth pathway, characterized by the structural transformation from a 65-type configuration (bilayer pentagon) to a 66-type configuration (bilayer hexagon) at the growth front of a hexagonal structure. Figure 3a shows a representative MD trajectory of this transformation, corresponding to the magenta path in Fig. 3d. Figure 3b,c demonstrates the cumulative spatial distribution of molecules in the top and bottom layers, respectively, sampled from all intermediate microstructures along the ordering pathway. These distributions confirm the general transition trends observed in the specific trajectory. The transition initiates with the adsorption of an ad-molecule ahead of the top-layer pentagonal structures, denoted by green circles in Fig. 3a–c. The ad-molecule induces distortions in the interlocked 2D bilayer pentagonal configuration, causing the breakage of the interlayer bonds and facilitating the in-plane adjustment of the bottom-layer molecules. As indicated by the red circles and arrows in Fig. 3a–c, the insertion of the additional molecule in the bottom-layer occurs from the side of the pentagon, which then reconnects with the top-layer molecule to form the bilayer hexagon.

A similar unlock-to-relock mechanism mediated by ad-molecules is observed at the zigzag edges of the growing ice islands, specifically during the transformation from a 665-type to a 666-type structures. As shown in Fig. 3f–h, ad-molecule insertion occurs at the top edge of the pentagonal site (indicated by green circles and arrows), including distortions in the interlocked bilayer configuration. This perturbation weakens the top-bottom connectivity, enabling the subsequently incorporation of a bottom-layer molecule from the lateral side of the pentagon. The additional bottom-layer molecule then reconnects with the top-layer molecule, completing the formation of a stable 666-type bilayer hexagon.

Although the ad-molecule-assisted growth mechanism at the dendrite tip growth and edge ripening is qualitatively similar, there are key differences in the crystallization barriers at these different growth sites, leading to fractal or compact morphology at varying temperatures. By measuring the potential energy of the molecules at tip and edge growth fronts, we find that the potential energy barrier ΔE for the

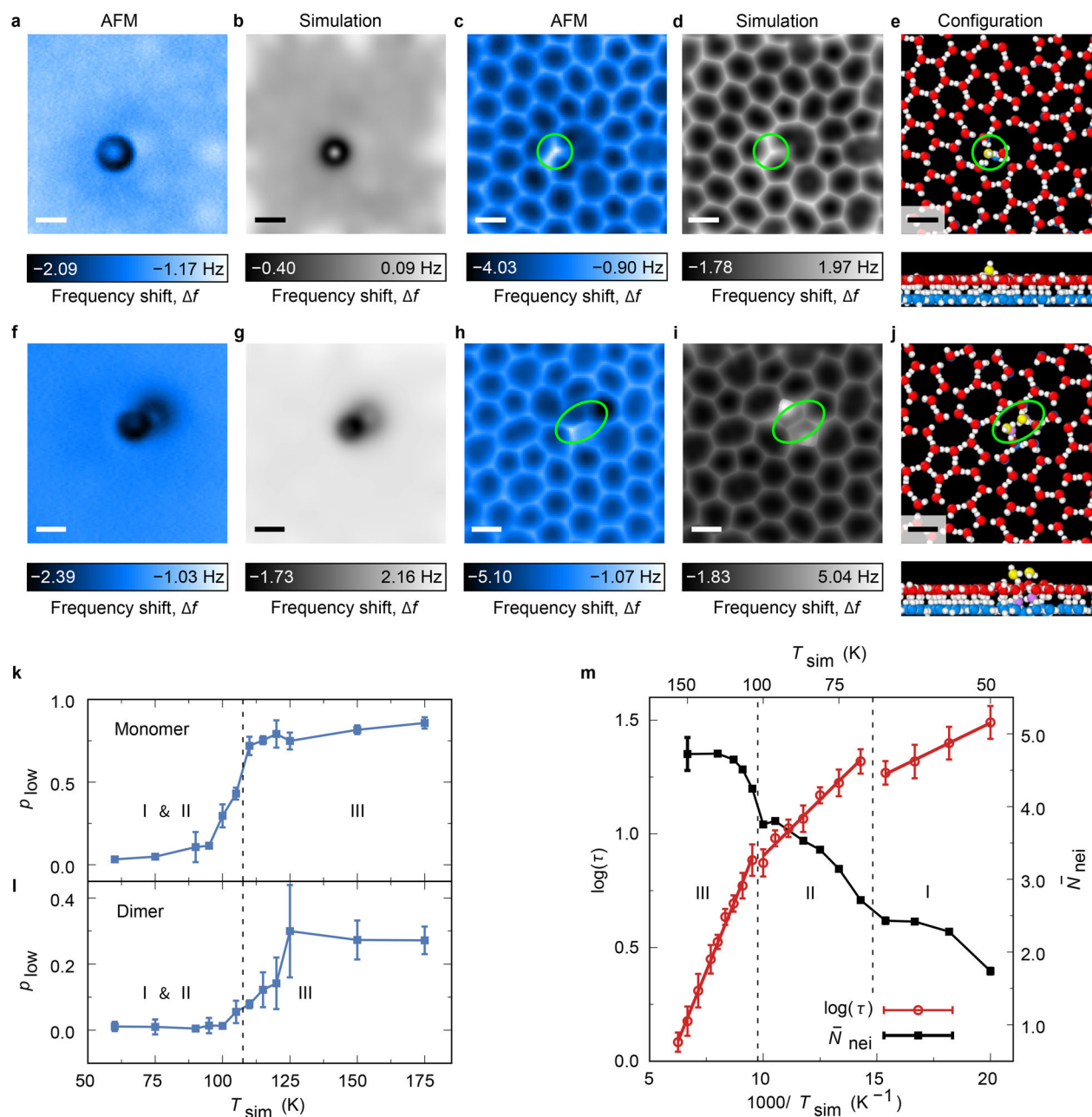


Fig. 2 | Microscopic structure and dynamic crossover of ad-molecules. a–j Constant height AFM and simulated images of adsorbed monomer (a–e) and dimer (f–j) above the 2D bilayer ice. The images resolved at larger (a, b, f, g) and smaller (c, d, h, i) tip heights clearly show the monomer and dimer structures marked by green circles. The atomic configurations are presented in e, j. The O atoms of the 2D top- and bottom-layers are colored in red and blue, while ad-molecules and directly underlying molecules are yellow and purple. The H atoms are white. Top and side views are displayed in the upper and lower rows, respectively. Tip heights: 200 pm (a), 23.10 Å (b), 20 pm (c), 20.80 Å (d); 300 pm (f), 24.00 Å (g), 0 pm (h), 20.80 Å (i). All scale bars are 5 Å. **k, l.** Occurrence probability p_{low} (marked by blue line and

squares) of underneath molecules below monomer (k) and dimer (l) configurations versus temperature from molecular dynamics (MD) simulations. Error bar represents six independent measurements. The vertical dashed line denotes the phase boundary between fractal (I & II) and compact (III) regimes. **m.** Arrhenius plot of the residence time τ (red circles in log scale with error bars) of ad-molecules as a function of annealing temperature in MD simulation. Error bar represents ten independent measurements. Vertical dashed lines denote dynamic crossovers consistent with the boundaries (I, II, III) identified in N_{nei} vs. T_{sim} relation (black line and dots same as Fig. 1g). Linear fits serve as guides to the eye.

zigzag edge growth from 665 to 666 (8.3 meV in Fig. 3j) is almost twice that for dendrite-tip growth from 65 to 66 (4.8 meV in Fig. 3e). This significant difference in energy barrier indicates that zigzag edge growth requires more thermal activation and exhibits a slower growth rate compared to dendrite tip growth. Furthermore, the excess number of amorphous states at the dendrite tip (represented by the bottom-left cluster in Fig. 3d) and the increased number of potential

sites for bond reconstruction contribute to a higher entropy and more possible routes for BHI crystallization compared with the zigzag edges. For instance, during zigzag edge growth, there is only one available site for the insertion of ad-molecules (green circles in Fig. 3f, g). In contrast, dendrite tip growth offers two symmetric potential sites (Fig. 3a, b). The availability of more possible insertion routes at the dendrite tip suggests a kinetic preference for dendrite tip growth over edge growth

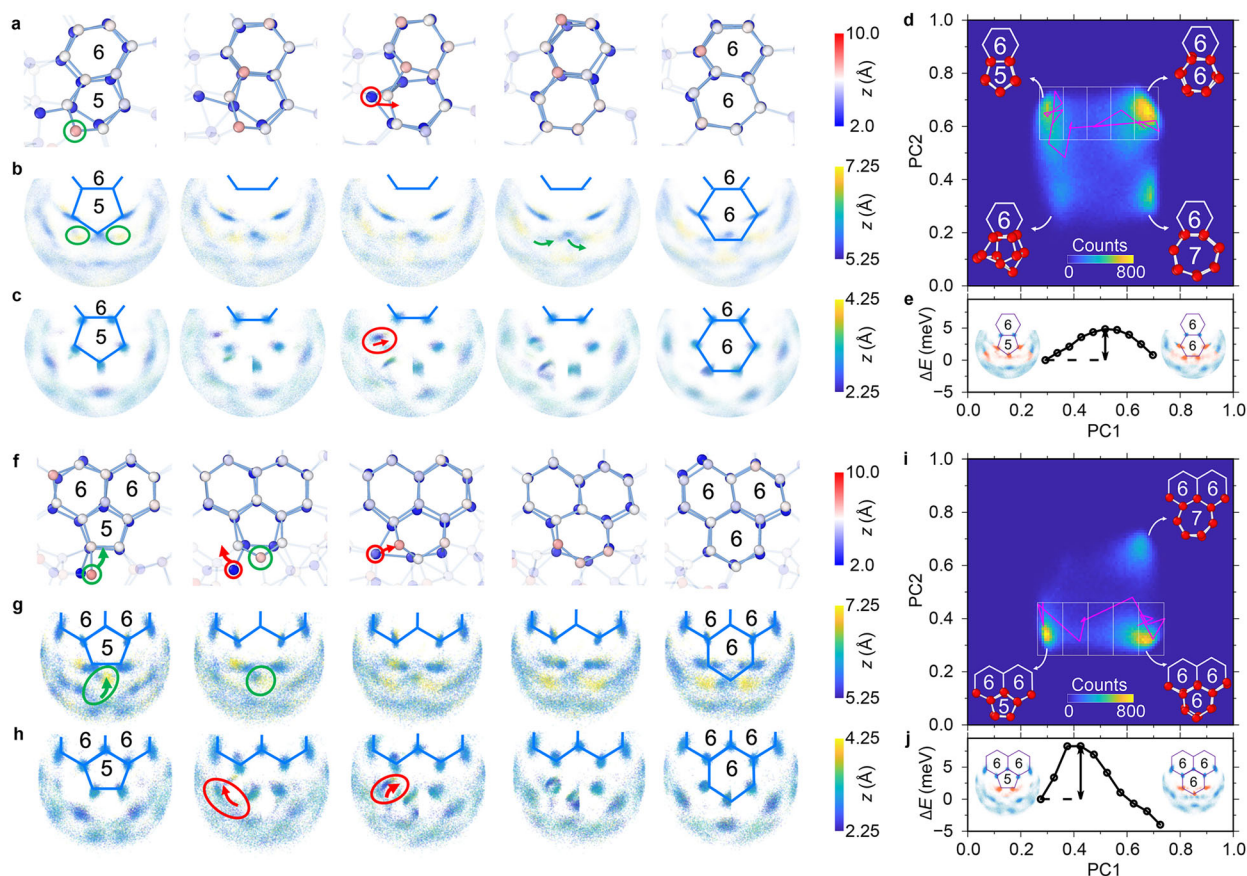


Fig. 3 | Microscopic ordering pathway at dendrite tip and zigzag edge. a–c, f–h Structural evolution of the dendrite tip growth from a 65- to 66-type structure (a–c) and the zigzag edge growth from a 665- to 666-type structure (f–h).

a, f demonstrate one typical trajectory of the time-lapse molecular dynamics (MD) snapshots during the growth from bilayer pentagonal defect to hexagonal crystal structures. **b, c** and **g, h** present the spatial distribution of the top (**b, g**) and bottom (**c, h**) layer molecules sampled along the 65 to 66 and 665 to 666 ordering paths (fenced in white boxes in **d, i**). The molecules are colored according to their perpendicular height z above the substrate. The incorporation of the 6-th molecules

into the 5-membered rings are circled and pointed by green and red arrows in the top and bottom layers. **d, i** Clustering of MD local configurations at dendrite tip (**d**) and zigzag edge (**i**) on the plane of two principal components PC1 and PC2 encoded by the machine learning algorithm. The color denotes the number counts of each local structure. The trajectories displayed in **a, f** are highlighted by the magenta pathways. **e, j** The potential energy per molecule ΔE as function of PC1. Both values are averaged over the growth front water molecules highlighted in orange as inset configurations. The zero energy baselines are set to be the potential energy of the 65- and 665-type structures.

during crystallization. Consequently, the free energy barrier for dendritic tip growth is markedly lower than that of compact edge ripening, which explains the macroscopic transition from fractal to compact structures observed in the phase diagram as the annealing temperature increases.

Kinetic pathway of 2D ice crystallization

To probe the kinetic pathway of 2D ice crystallization, we explore the temporal structural evolution of BHI crystallization through MD simulation. Specifically, we examine the time dependence of the normalized size of the largest BHI island L_m ⁴³ (see Methods) and the fractal dimension D_f during the crystallization process. We investigate two typical crystallization trajectories: one at a medium annealing temperature $T_{\text{sim}} = 100$ K of fractal phase II regime (Fig. 4a), and another at a high temperature $T_{\text{sim}} = 175$ K in the compact phase III regime (Fig. 4d), to uncover the underlying crystallization pathway.

At the medium annealing temperature of 100 K (Fig. 4a), the BHI islands grow slowly in a dendritic fashion, maintaining a fractal dimensionality $D_f \approx 1.7$. The evolution of the largest BHI island exhibits a sudden increase at $t = 380$ ns, corresponding to the connection of two fractal islands (colored in red and pink in Fig. 4b) into a larger cluster (red in Fig. 4c). Due to the insufficient thermal excitation at this temperature, the amorphous 2D ice crystallizes only into localized

fractal islands via dendrite tip growth, without forming fully connected compact structures.

In contrast, at a higher temperature of 175 K, the BHI islands exhibit a rapid growth rate, with successive increase in the largest crystal size and fractal dimension, indicating a two-step crystallization mechanism² (see Fig. 4d and Supplementary Fig. 15 for other temperatures). In the first step, the BHI islands grow along the fractal branches and merge into a percolated cluster, characterized by the growth of the largest crystal size L_m toward 1. A typical configuration of the largest percolated BHI cluster is shown in red in Fig. 4e, spanning the simulation box in a fractal morphology. In the second step, the BHI clusters undergoes a fractal-to-compact transformation, as the dendrite branch ripens through zigzag edge growth to form compact islands, with the fractal dimension D_f gradually increasing to 2.0. The final compact BHI islands are shown in Fig. 4f. This two-step growth mechanism^{44,45}, characterized by fractal-percolation followed by fractal-to-compact transition, stands in stark contrasts to the classical nucleation theory featured with a critical nucleus, thereby provides substantial evidence for a non-classical crystallization pathway at atomic level.

To further investigate the kinetic ordering pathway of the two-step growth, we compute the free energy surface (FES) of 2D ice on graphite substrate using metadynamics simulations^{46,47} at $T_{\text{sim}} = 175$ K

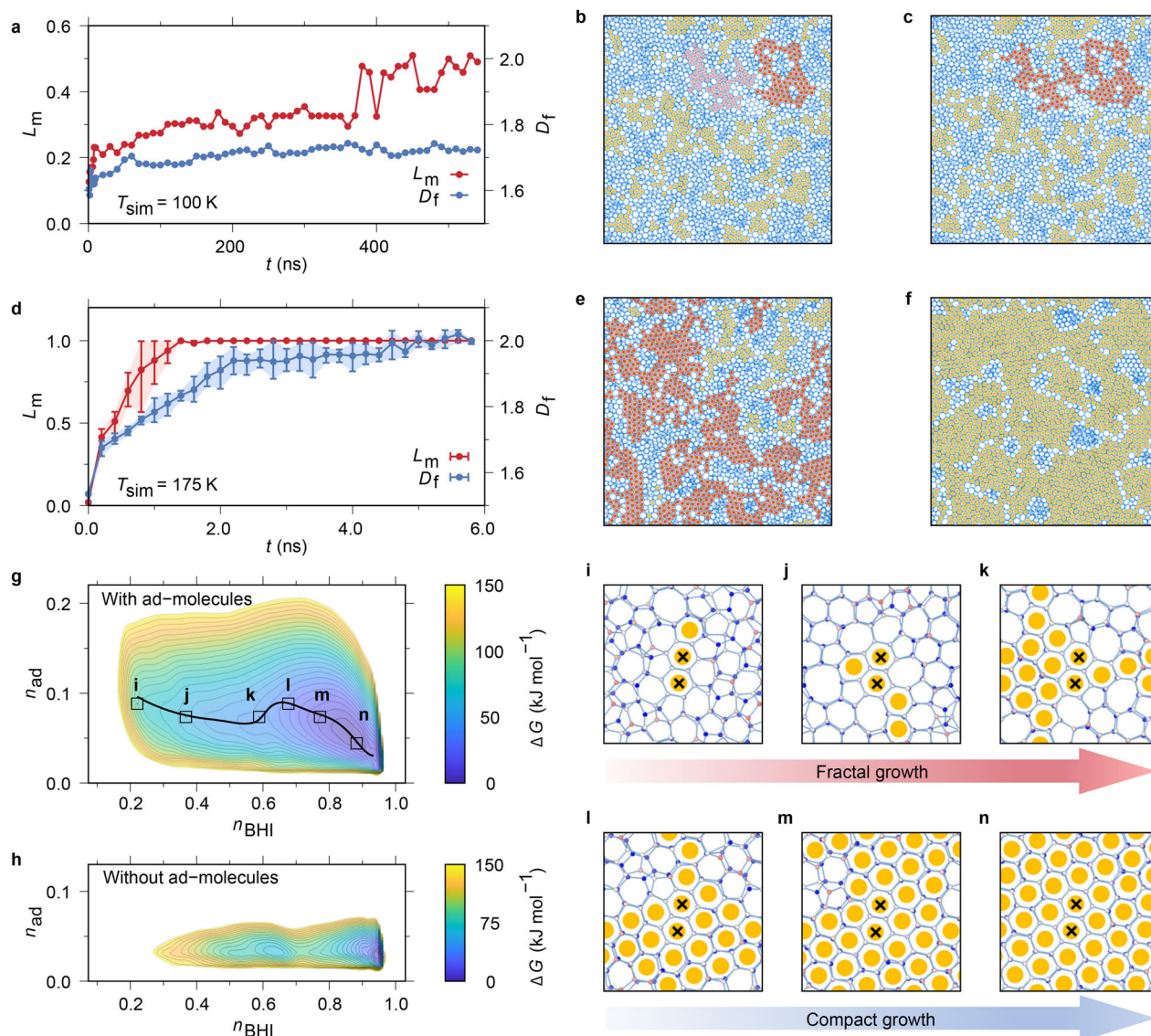


Fig. 4 | The two-step crystallization mechanism of 2D ice. **a, d** Temporal evolution of the largest crystal size L_m (red) and fractal dimension D_f (blue) at **(a)** $T_{sim} = 100$ K and **(d)** 175 K, using one single trajectory of molecular dynamics (MD) simulation in **a** and averaging over five independent trajectories in **d**, respectively. The upper (lower) bounds of the error bars represent the maximum (minimum) values obtained from five independent measurements. **b, c** MD snapshots of the bilayer hexagonal ice (BHI) islands before ($t = 370$ ns in **b**) and after ($t = 380$ ns in **c**) the connection of two fractal islands at 100 K in **a**. The two islands are marked by red and pink in **b**, and the combined island is highlighted in red in **c**. The rest of BHI

islands are colored in yellow. **e, f** MD snapshots of the percolated BHI island (red) after fractal growth as L_m reaches 1 (**e**), and compact BHI islands after compact growth (**f**) at 175 K. **g, h** Free energy surfaces of 2D ice at 175 K, in terms of BHI molecule fraction n_{BHI} and ad-molecule fraction n_{ad} , on the graphite surface without confinement (**g**) and confined between two parallel surfaces (**h**). **i–n** The representative structures sampled along the minimum free energy path of crystallization (black line in **g**). The black crosses mark the fixed BHI seeding nucleus, and the interlocked hexagons grown from the seed are marked in yellow disks. The molecules are colored from blue to red with increasing height above the substrate.

(see Methods). Two characteristic order parameters are employed in enhanced sampling: the number fraction n_{BHI} of BHI molecules grown from a fixed BHI seed (black crosses in Fig. 4i–l), and the number fraction n_{ad} of ad-molecules above the bilayer ice layers (see Methods and Supplementary Fig. 16). As shown in Fig. 4g, the minimum free energy pathway^{8,45} of BHI crystallization (black curve) undergoes a gradual decrease in ad-molecule numbers at initial stage, with $n_{BHI} = 0.2 - 0.6$, followed by an increase in n_{ad} before reaching the global minimum of fully crystallization. The dynamic variations in the number of ad-molecules directly illustrate their participation in the 2D crystallization process. The facilitating role of the ad-molecules is further validated by estimating the FES of confined 2D ice between two parallel surfaces that restrict the appearance of ad-molecules (Fig. 4h and Supplementary Fig. 16e). The resulting FES under identical

temperature shows significant energy barriers, particularly during the second step at $n_{BHI} \approx 0.7$, emphasizing the surfactant role of ad-molecules in reducing the energy barrier of 2D ice crystallization.

A typical two-step mechanism is thus revealed through the sequence of configurations in Fig. 4i–n. In the first stage of fractal growth in (i $\rightarrow$ j $\rightarrow$ k), the bilayer hexagons primarily grow at the dendrite tips and form a percolated cluster surrounded by interlocked bilayer pentagons and heptagons. During this stage, the BHI islands maintain a dendritic form, with a fractional dimension $D_f = 1.72$ (see Supplementary Fig. 17). In the second stage (l $\rightarrow$ m $\rightarrow$ n), compact growth occurs with the breakage and rearrangement of interlocked bonding network assisted by ad-molecules. During this stage, the fractal dimension D_f increases from 1.72 to 2.08 (see Supplementary Fig. 17) until fully crystallization. This observation elucidates the non-

classical crystallization pathway from the perspective of energy landscape.

Discussion

Our direct imaging of the intermediate microstructures during 2D ice crystallization provides experimental evidence—at atomic resolution—for a non-classical crystallization pathway where critical nucleus formation is absent. This highlights the anisotropic nature of the crystallization process, driven by the kinetic bias of dendrite tip growth and percolation with lower energy barriers. As annealing progresses, compact growth occurs at the dendrite edges, overcoming higher energy barriers to yield compact crystalline domains at elevated temperatures.

Given the structural incommensurability (see Supplementary Fig. 2) and weak interaction between water and the carbon surface³⁵, this unconventional fractal-to-compact crystallization mechanism may not be limited to graphite but can extend to a broader class of hydrophobic substrates^{35,48}, as supported by MD simulation of 2D ice crystallization on a structureless Lennard-Jones 9-3 surface³⁶ (see Supplementary Fig. 18). It may also apply to confined geometries where 2D ice forms under ambient conditions⁴⁹. Similar fractal-like morphologies have been observed in water confined between 2D materials^{50,51}, suggesting that our findings may provide atomic-scale insight into these mesoscale growth phenomena. Moreover, such a non-classical ordering pathway is also reproduced in the simulation of a 2D Lennard-Jones systems (Supplementary Fig. 19), implying that the presence of multiple metastable, pre-ordered states may be a general feature of 2D crystallization^{52,53}. Additionally, the autocatalytic role of ad-molecules in lowering the energy barrier during compact ripening provides key insights into the 2D growth process. Unlike previous studies³⁸ focused on 2D in-plane behaviors, our findings highlight the significant impact of the out-of-plane reconstructions on 2D crystallization embedded in a 3D space. These insights may enable atomic-level control of a wide range of 2D materials, including 2D silica glasses^{21,22} and transition metal dichalcogenides⁵⁴, shedding lights on the design and control of crystallization pathways in low-dimensional materials.

Methods

AFM experiments

All the experiments were performed with combined noncontact AFM/STM systems from CASAcme (China), by using a home-made qPlus sensor^{25,30–32} equipped with a tungsten (W) tip (spring constant, $k_0 \approx 1800 \text{ N m}^{-1}$; resonance frequency, $f_0 = 30.4 \text{ kHz}$; quality factor, $Q \approx 100,000$). The highly oriented pyrolytic graphite sample is purchased from MaTeck. To resolve the atomic structures of 2D ice, the samples are rapidly transferred from the annealing temperature to a low temperature of 5 K for AFM imaging under ultrahigh vacuum ($<3 \times 10^{-10} \text{ mbar}$). The AFM frequency shift (Δf) images were obtained with the CO-functionalized tips in constant-height modes. The oscillation amplitude of AFM imaging is 200 pm if not specifically mentioned. The CO-tip was obtained by positioning the tip over a CO molecule on the Au (111) surface at a set point of 100 mV and 10 pA, followed by increasing the current to 400 pA. Then the CO-tip is carefully transferred to graphite surface for further imaging. The weakly perturbative imaging of the metastable transient states during crystallization is achieved by probing the high-order electrostatic force between the CO-tip and water molecules³². Image processing was performed by Nanotec WSxM⁵⁵.

Identification of atomic structures with ad-molecules

We implemented ML-aided structure identification algorithm⁴⁰ to reconstruct the 3D structure of the 2D ice with ad-molecules from experimental AFM images. The key steps are summarized as follows:

Firstly, the ML algorithm⁴⁰ is implemented to predict the bilayer ice structure. The neural network took experimental AFM images with varying tip heights as input and predict the 3D position of the top layer of 2D ice. Given the buckled nature of 2D ice, the lower layer structure of BHI was reconstructed by mirroring the top layer.

Secondly, the ice configuration predicted by ML algorithm is refined using MD simulations. The predicted configuration was initially placed on a large graphene sheet and a 5 nm vacuum slab was added in z direction. Periodic boundary conditions were applied in all three directions of the simulation box. The system was relaxed using MD simulations in the NVT ensemble at 150 K for 1 ns. During relaxation, the graphene substrate and the relative positions of O atoms were fixed, allowing the 2D ice to move as a whole while H atoms adjusted freely to refine the H-bonding network. The Tip4p/Ice⁵⁶ force field was used for describing water-water interactions. The C atoms of graphene interact with O atoms of water through Lennard-Jones interactions with distance parameter $\sigma = 3.3584 \text{ \AA}$ and interaction strength $\epsilon = 0.121486 \text{ kcal mol}^{-1}$. After relaxations, the monomer and dimer molecules were manually placed on top of the 2D BHI structures. We further ran series simulations with different number of water molecules by deleting or adding molecules at the bottom layer of BHI to obtain stable structures. A spring force constant $k = 0.5 \text{ kcal mol}^{-1} \text{ \AA}^{-2}$ was applied to keep the ad-molecules in their initial positions, while graphene and the BHI boundaries were fixed. The final resolved configurations were proved to be stable in MD simulations.

Finally, the validity of the refined configuration was further confirmed through DFT calculations, ensuring it represented a low-energy state. We also simulate the AFM image of the refined configurations using the probe particle method. This simulated AFM image is then directly compared with the original experimental AFM image. Good agreement between simulated and experimental AFM images confirms the validity of the structure.

DFT calculation

DFT calculations were performed using the Vienna Ab initio Simulation Package (VASP version 6.2)^{57,58}. Projector-augmented wave pseudo-potentials were used with a cutoff energy of 550 eV for the expansion of the electronic wave functions⁵⁹. Van der Waals corrections for dispersion forces were considered using the 'optB86b-vdW' functional^{60,61}. In our DFT calculations, a two-dimensional ice layer containing either a monomer or a dimer is placed on top of a graphene layer. The lattice constant for graphene was set to $2.460 \text{ \AA}$. The graphene layer was kept fixed during the calculations. The simulation box size for both structures containing either a monomer or a dimer was $34.09 \text{ \AA} \times 34.44 \text{ \AA} \times 35 \text{ \AA}$. The water molecules adjacent to the edge of the AFM image were arranged to replicate the experimental conditions, with their positions fixed. The majority of water molecules were not adjacent to the edge, and all of these were allowed to relax, including those surrounding the monomer or dimer. For all these calculations, only the Γ point was used and the thickness of the vacuum slab in z direction was larger than $15 \text{ \AA}$. The geometry optimizations were carried out at fixed volume with a force criterion of $0.02 \text{ eV \AA}^{-1}$.

Simulations of AFM images

The Δf images were simulated using the probe particle model^{62,63}, where a flexible CO probe is connected to the tip by harmonic springs in radial and lateral directions. The probe relaxes in response to local surface forces, mimicking high-resolution AFM measurement^{25,32,64}. The following parameters were used for the probe-particle tip model in our simulations: the effective lateral stiffness $k = 0.75 \text{ N m}^{-1}$ and effective atomic radius $R_c = 1.661 \text{ \AA}$. Note that R_c differs from the cutoff radius ($r_c = 3.5 \text{ \AA}$) of the bonding distance between water molecules in MD simulations. A quadrupole-like charge distribution at the tip apex was used to simulate the CO-tip with $q = -0.10 \text{ e}$ (e is the elementary charge, and q is the magnitude of quadrupole charge at the tip apex).

The electrostatic potentials employed in AFM simulations were obtained from DFT calculations. Parameters of the Lennard-Jones pairwise potentials for the C, O and H atoms used in AFM simulations are: $\sigma_C = 1.908 \text{ \AA}$, $\epsilon_C = 3.729 \text{ meV}$, $\sigma_O = 1.661 \text{ \AA}$, $\epsilon_O = 9.106 \text{ meV}$, $\sigma_H = 1.487 \text{ \AA}$ and $\epsilon_H = 0.681 \text{ meV}$. The tip height in the AFM simulations is defined as the vertical height of the metal tip apex. The oscillation amplitudes of the simulated AFM images are 200 pm.

MD simulations

The MD simulations are carried out using the LAMMPS package⁶⁵ and the monoatomic mW model⁶⁶ is implemented to describe the water-water interaction. A periodic graphene layer is used to model the carbon substrate³², which is kept rigid during the simulations. The water-carbon interaction is simulated by a two-body Stillinger-Weber potential with $\sigma = 0.32 \text{ nm}$ and $\epsilon = 0.13 \text{ kcal mol}^{-1}$ for the length and energy parameters³⁶, which corresponds to a contact angle of 86° consistent with the experimental results. A total number of 8997 water molecules (≈ 1.06 perfect bilayer hexagonal ice molecules) are placed on the substrate with an area of $204.192 \text{ \AA} \times 201.396 \text{ \AA}$. We also examine other coverage ratio containing 8277 and 9594 water molecules (see Supplementary Fig. 20) to verify the robustness of our results. The Nosé-Hoover thermostat is used to maintain the temperature with a time step of 1 fs. The initial amorphous state is constructed from a liquid film equilibrated at $T = 350 \text{ K}$ and then quickly quenched to $T = 10 \text{ K}$ to freeze the disordered morphology. This fast-quenching strategy can be more efficient than a deposition simulation, and there is no essential difference in the overall structural configurations of the amorphous solids formed through fast-quenching and deposition¹³. Then the amorphous water film is annealed at constant temperatures ranging from 50 to 200 K for 10^9 MD time steps to assess the stability of the 2D ice structures.

Free energy surface calculation

Well-tempered metadynamics (WTMetaD) simulations are carried out using LAMMPS patched with PLUMED package⁶⁷ to calculate the free energy surface. A simulation cell with dimensions of $L_x \times L_y = 24.68 \text{ \AA} \times 25.65 \text{ \AA}$ is used to perform the enhanced sampling simulation, containing one layer of perfect BHI (see Fig. 4n). A BHI seeding nucleus composed of two hexagonal rings (black crosses in Fig. 4i-n) is fixed to converge the free energy surfaces more efficiently since the system never fully melts⁶⁸, and the newly formed BHI grown from the nucleus remains in an identical crystal orientation. Two collect variables (CV) are implemented to characterized the number fraction of 2D BHI n_{BHI} and ad-molecules n_{ad} during the simulation. The BHI molecules are distinguished from amorphous water molecules based on the environmental similarity parameter⁶⁹,

$$k_{\chi_0}(\chi) = \int d\mathbf{r} \rho_{\chi}(\mathbf{r}) \rho_{\chi_0}(\mathbf{r}), \quad (1)$$

which measure the difference between the examined molecule environment χ and χ_0 of a perfect BHI crystal structure. Here $\rho_{\chi}(\mathbf{r}) = \sum_{i \in \chi} e^{-|\mathbf{r}_i - \mathbf{r}|^2 / 2\sigma^2}$ denotes the local density that sums over the neighbors in the environment χ and is broadened by a Gaussian function. More details are available in ref. 69. The broadening parameter is set as $\sigma = 0.055$ to yield the optimal classification of BHI and amorphous environments (see Supplementary Fig. 16c). Note that the BHI lattice has four equivalent environments with different orientations, so the order parameter is defined as the closest similarity between the examined environment and any of the four environments in BHI.

$$k_{\text{BHI}}(\chi) = \max\{k_1(\chi), k_2(\chi), k_3(\chi), k_4(\chi)\}, \quad (2)$$

The ad-molecules are defined as the adsorbed molecules with vertical height $z > 7 \text{ \AA}$ above the graphite substrate, according to the transverse density profiles shown in Supplementary Fig. 16f. After introducing the two CVs, the WTMetaD bias is constructed by depositing Gaussians with a height of 2 kJ mol^{-1} every 1 ps. The bias factor of the WTMetaD ensemble is set to 100.

ML-based clustering of local configurations

A neural network-based autoencoder⁷⁰ is used to analyze the intermediate micro-structures and ordering pathways at the BHI growth front. Firstly, the local configurations at the dendrite tip and zigzag edges are separately extracted and aligned from MD trajectories. Then we define 12 pivot points $\{r_1(p), r_2(p), \dots, r_{12}(p)\}$ in representation of the 5-, 6-, 7-membered ring structures ahead of the growth front. The shortest distances between the molecules in the examined environment χ and the pivot points are computed as a 12-dimensional vector to characterize the local environment, namely $F(\chi) = (\min\{\|r_i(\chi) - r_i(p)\|\})$ with i goes through 1 to 12. The dimensionality of the vector is then lowered into two dimensional features PC1 and PC2, using a ML-based autoencoder⁷¹ neuron network, as is shown in Fig. 3b, e. More details of the clustering algorithm are given in Supplementary Fig. 13.

Residence time of ad-molecules in simulation

The residence time τ is defined as the reciprocal of hopping frequency between the 3D adsorbates and 2D ice layers (see Supplementary Fig. 12b). To measure the hopping frequency, we count the total number of hopping events N_h in 1 ps with an ultrafast sampling rate of 100 frames per ps. The hopping event refers to the drop of an ad-molecule from the 3D adsorbed layer ($z > 6.65 \text{ \AA}$) to the 2D bilayer ($z < 6.65 \text{ \AA}$), representing the annihilation of an ad-molecule. The residence time is defined as $\tau = (N_h/N_{\text{ad}})^{-1}$ in units of ps, which is normalized by the total number of ad-molecules N_{ad} . Several independent sampling of τ are conducted at different time steps during the crystallization process (see Supplementary Fig. 12a). The temperature-dependent residence times at different phase regimes are fitted by an Arrhenius relation $\tau = \tau_0 \exp[E_A/(k_B T_{\text{sim}})]$ ⁴¹.

Normalized size of largest BHI island

The normalized size of largest BHI is computed as $L_m = \frac{1}{2} \max(l_x/L_x + l_y/L_y)$ ⁴³. Here l_x and l_y are the spanning length of the largest BHI island along the planar x and y dimensions, normalized by the lateral box sizes L_x and L_y . At $L_m = 1$ the largest BHI island spans across both the x and y dimensions of the MD simulation box.

Data availability

The data that support the findings of this study are available from the corresponding authors upon request. Source data for Figs. 1-4 and Supplementary Figures are provided via Zenodo⁷².

Code availability

The simulation codes used in this study are available from the corresponding authors upon request.

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Author contributions

L.X., Y.J., and E.-G.W. designed and supervised the project. Y.T., Z.Y., D.G., and J.G. performed the AFM experiments. Z.-F.Y., B.T. and T.L. performed the MD and DFT simulations. Z.-F.Y., Y.T., B.T., T.L., C.-H.L., L.X., Y.J., and E.-G.W. analyzed the data. Z.-F.Y., Y.T., B.T., T.L., L.X., Y.J., and E.-G.W. wrote the manuscript with the inputs from all other authors.

Competing interests

The authors declare no competing interests.

Additional information

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